Toss Life

Overview

Toss Life is a multiplayer life-simulation game where students live a whole financial life, from their first earnings to retirement, and learn by making real money decisions with virtual money.

Tagline: Life tosses the coin. Your choices decide where it lands.

The problem

- Most Grade 11 and 12 students will soon earn, borrow and invest, yet few have been taught how money grows or shrinks.
- Many families believe a bank FD is the only safe place for money, without seeing how inflation reduces its real value.
- Young people are prime targets for loan apps, fake job offers and "double your money" schemes.
- Information is available on Google, but **reading about money does not change behaviour. Experiencing consequences does.**

Why a game

- **Life is the biggest teacher of money.** Toss Life compresses 40 years of life into a few hours, so students see consequences that normally take decades.
- **Mistakes are free.** Students can lose virtual money, get scammed, or borrow too much, and learn without real harm.
- **Friends make it stick.** Lending, borrowing and bidding with classmates makes interest and risk personal and memorable.
- **Investing is the core.** Every chapter builds toward one big idea: time, patience and diversification grow wealth more than keeping cash or relying only on FDs.

How the game works

Every chapter follows the same simple loop, so students learn the controls once and then focus on decisions.

1. **Earn.** The chapter begins with income: a stipend, a salary or business earnings.

2. **Get tasks.** A to-do list with due dates: pay the phone bill, pay college fees, attend a friend's wedding, save for a laptop, build an emergency fund.
3. **Decide.** For every rupee, choose to spend it, keep it, invest it, or lend it to a friend. Any amount, no single right answer.
4. **Toss.** Time fast-forwards. A coin toss decides each year's luck: a good or bad market year, a bonus or a medical bill. Luck is random; how much it hurts or helps depends on the player's earlier choices.
5. **Review.** The chapter ends with a Life Report showing what went well, what could be better, and a "What if" replay of a smarter path.

The toss: the signature mechanic

- The coin toss is a visible, dramatic moment that every player waits for.
- It teaches that **no one controls luck, but everyone controls preparation.** A player with an emergency fund survives a bad toss; one without it is forced to borrow.
- Market years are tossed too. Over many tosses, students see that a long-term SIP smooths out bad years, while trying to time the market often loses.

Money Mindset Check: 6 indirect questions

Before playing, students answer 6 quick, everyday questions. None asks about income, family, caste, location or any personal data, yet together they reveal how the student thinks about money and how much they already know.

Each question has 3 or 4 tap-to-answer options with pictures, and takes about 10 seconds.

#	Question	Options	What it reveals
1	Your friend's birthday is next week. Would you spend ₹500 on a gift?	Yes, gifts matter / I'd spend less or make something / Only if I had saved for it	Spending style: impulsive or planned
2	You win ₹1,000 in a contest. Take it today, or ₹1,300 after 6 months?	Today / After 6 months / Depends on what I need	Patience, the key to compounding

#	Question	Options	What it reveals
3	At a fair, pay ₹100 to toss a coin. Heads you get ₹250, tails nothing.	Play once / Skip it / Keep playing until I win	Risk appetite
4	Your grandparents say a samosa cost ₹2 when they were young. Why is it ₹20 now?	Shops got greedy / Prices rise over time / Not sure	Understanding of inflation
5	A relative says, "Keep all your money in an FD, it's the safest." You think:	Fully agree / Safe but grows slowly / Depends on the goal / What's an FD?	Investment knowledge
6	A message says: "You won ₹10,000! Pay ₹99 to claim it." You:	Pay, it's only ₹99 / Ask a friend first / Ignore and report	Scam awareness

How answers are used

- Questions 1 to 3 give a **Mindset score** (how the student behaves with money).
- Questions 4 to 6 give a **Knowledge score** (what the student understands).
- The two scores place the student into one of four personas.
- **The same 6 questions return at the end of the game.** The change in answers is a direct, privacy-safe measure of learning.

The four personas

Personas come from two scores: how the student behaves with money (Mindset) and what they know (Knowledge). Every persona is framed with a strength and a blind spot, never as "good" or "bad."

Persona	Mindset	Knowledge	Strength	Blind spot	How the game adapts
Moment Maker	Spends now	Beginner	Generous, enjoys life	Little saved for surprises	Starts Chapter 1 with a guided month; more budgeting tasks and surprise expenses

Persona	Mindset	Knowledge	Strength	Blind spot	How the game adapts
Safe Keeper	Plans ahead	Beginner	Disciplined saver	Keeps everything in cash or FDs	Starts Chapter 1; inflation and Real Value are shown early to push beyond FDs
Thrill Seeker	Takes risks	Some knowledge	Bold, open to investing	Chases quick returns	Starts Chapter 1; more scam offers and volatile markets; diversification is the focus
Smart Planner	Plans ahead	Aware	Patient and informed	Can over-plan and miss opportunities	Can start Chapter 2; advanced options and bidding unlock sooner

Why this works

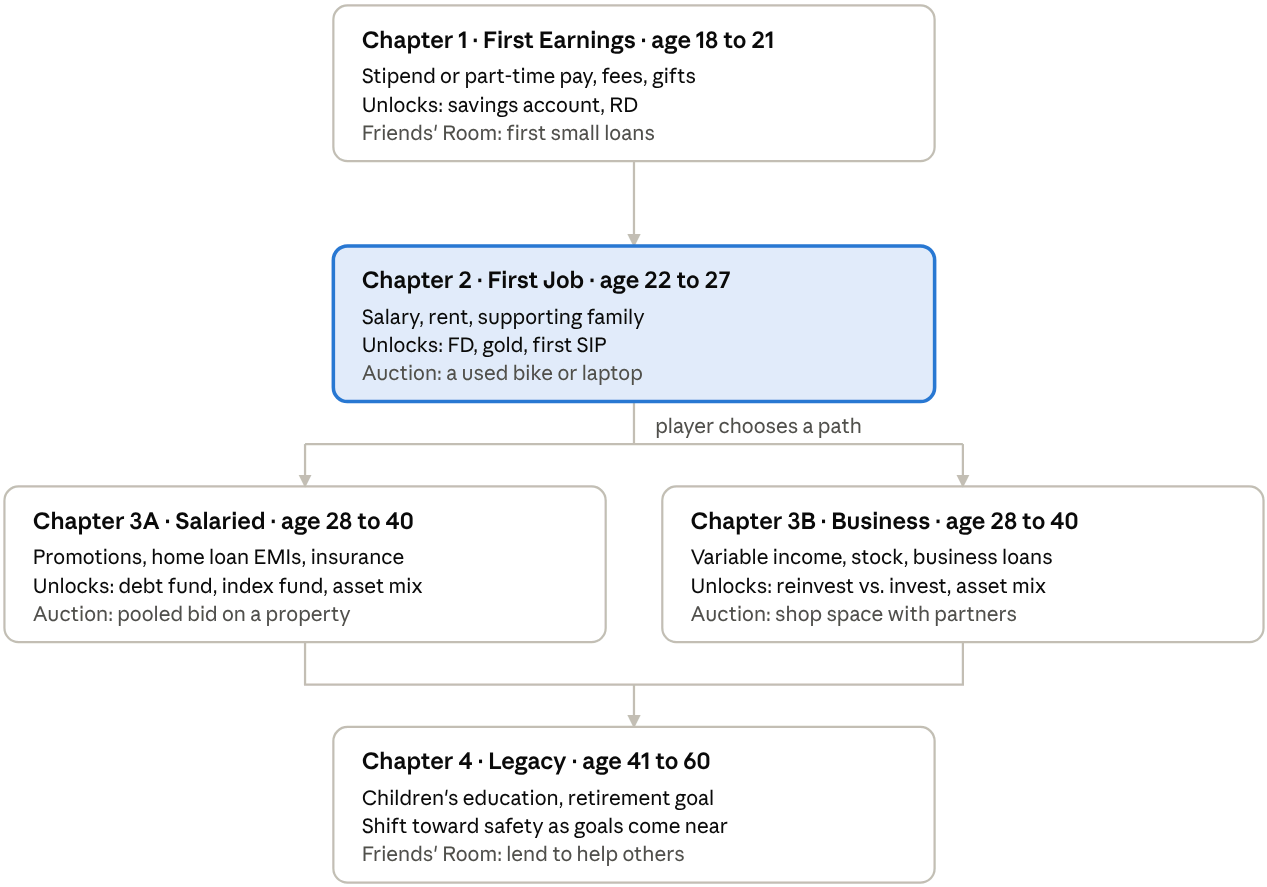
- **Only 4 personas, one game engine.** A persona changes starting point, mentor tips, difficulty and the mix of events, not the whole story. New personas can be added later as new settings.
- **Persona can change.** If a Moment Maker starts saving well, the game recognises it and shifts their persona. Growth becomes visible.
- **Students can try other personas.** Replaying as a different persona shows how the same life can turn out very differently.

Life chapters

Every player climbs the same ladder of life stages, and at Chapter 3 chooses a career path, because a salaried employee and a business owner make very different money decisions.

Every player climbs one ladder; at Chapter 3 they choose their path

Highlighted chapter is fully playable in the hackathon prototype



Life Ladder · 4 stages, 5 chapters, 2 career paths

- Each chapter covers more years than the last, so compounding becomes more visible as players climb.
- New investment options and bigger multiplayer opportunities unlock at each stage.
- Players can replay Chapter 3 to try the other path and compare outcomes.

Investing: the heart of the game

Toss Life does not say "mutual funds always beat FDs." It lets students discover that the right choice depends on the goal and how long they can wait, and that over long periods, diversified investing and SIPs usually grow wealth more than cash or FDs alone.

Investment options unlock chapter by chapter

Option	Unlocks	Assumed yearly return in game	Lesson it teaches
Cash	Chapter 1	0%	Inflation quietly eats idle money
Savings account	Chapter 1	~3%	Good for emergencies, not for growth
Recurring deposit (RD)	Chapter 1	~6.5%	Saving as a monthly habit
Fixed deposit (FD)	Chapter 2	~6.5 to 7%	Safe and fixed, but slow after inflation; penalty if broken early
Gold	Chapter 2	~8% average, swings	Not as safe as people assume
SIP in a diversified equity fund	Chapter 2	~11% average, some years negative	Time and regular investing beat timing the market
Debt fund	Chapter 3	~7%, small swings	A middle ground between FD and equity
Index fund	Chapter 3	Follows the market average	Low cost, broad diversification
Asset mix	Chapter 3	Blend of the above	Diversification protects against bad tosses

All rates are illustrative assumptions, to be confirmed with Vanguard mentors. No real fund, bank or company names are used.

Five investment techniques the game teaches

1. **Compounding:** returns earn returns. A "growth tree" visual grows each year the money stays invested.
2. **SIP and rupee-cost averaging:** investing a fixed amount monthly buys more units in bad years, smoothing out the tosses.
3. **Diversification:** spreading money across options means a bad toss hits only part of it.

4. **Risk vs. time:** the longer the goal, the more risk a player can afford. Short-term money stays safe; long-term money can grow.
5. **Goal-based investing:** the emergency fund comes first, then short-term goals in safe options, then long-term goals in growth options.

The FD myth-buster: Real Value

Every investment shows two numbers:

- **Amount:** how much money you have
- **Real Value:** what that money can actually buy after prices rise

When students see an FD balance grow but its Real Value barely move, while their SIP's Real Value has grown, the belief that "FD is always best" changes on its own.

Multiplayer: lending, interest and bidding

In the Friends' Room, 3 to 6 classmates play their lives side by side. They can lend to each other, borrow, and bid for opportunities, which turns interest, credit and risk into real experiences between friends.

Students join a room with a short code. When friends aren't available, computer-controlled friends fill the room, so the game also works solo.

1. Lending and borrowing

- Any player can **request a loan** when short of money, for example after a bad toss.
- Friends can **offer a loan** and set the interest rate and repayment date themselves.
- The borrower compares offers side by side: a friend at 2% a month, the in-game bank at a yearly rate, or a loan app with very high charges.
- On repayment, the borrower pays back the loan plus interest. The game shows exactly how much extra was paid.
- Late or missed payments lower the borrower's **Trust Score**, which works like a real credit score: a low score means higher interest or no loan next time.
- Lenders face real risk too: if a friend doesn't repay, the lender loses money and learns why lending needs care.

Concepts taught: simple and compound interest, the cost of borrowing, credit score, lending risk, and why high-interest loans are dangerous.

2. Bidding for opportunities

Each chapter, an **Opportunity Auction** appears, such as a used bike at a discount, a small shop space, or a share in a friend's business idea.

- Players bid against each other with sealed or open bids.
- Bidding too high leaves too little for tasks; bidding too low misses the chance. Students learn that **price and value are not the same.**
- **Pool Bid:** friends can combine money to bid together and share the returns by how much each put in. The game then reveals: *"This is exactly how a mutual fund works. Many people pool money, it is invested together, and returns are shared."*

Concepts taught: value vs. price, opportunity cost, pooled investing and how mutual funds work, sharing risk and reward.

Why multiplayer matters

- Lessons from a friend charging you interest are remembered far longer than a definition.
- It creates natural classroom discussion: who lent, who borrowed, who won the auction and was it worth it.
- It builds trust and responsibility, skills that go beyond money.

Scams and life events

Scams appear the way they do in real life: inside the investment list, as messages, and as tempting offers right after a bad toss, when players are most likely to fall for them.

Scams by chapter

Chapter	Scam	Red flags taught
1	"You won a prize, pay a small fee to claim"	Paying to receive money
1	Job offer that asks for a registration fee	Real jobs don't charge to hire
2	Call asking for OTP to "unblock" a bank account	Banks never ask for OTP or PIN
2	Instant loan app during an emergency	Hidden charges, misuse of phone data

Chapter	Scam	Red flags taught
3	"Guaranteed 5% a week" tips group and trading app	Guaranteed high returns, unregistered platforms
3	A friend's "double your money" scheme in the Friends' Room	Returns paid from new members' money

If a player falls for a scam:

1. The money is lost, clearly shown.
2. A **Red Flags card** points out the warning signs that were in the offer.
3. The same type of scam returns later in a new disguise. Spotting it earns the **Scam Buster** badge.

Life events from the toss

- **Bad tosses:** medical emergency, phone breaks, job loss for a few months, a family wedding
- **Good tosses:** a bonus, a scholarship, a promotion. These test whether the player saves a windfall or spends it all.
- **Market tosses:** good and bad years for market-linked options

Life Report, scoring and progress

Progress depends on a Financial Health Score, not just total money, so students learn that balanced decisions win, not reckless ones.

Financial Health Score (out of 100)

Part	Points	What it checks
Tasks on time	25	Bills, fees and needs paid by their dates
Safety net	20	An emergency fund that survived a bad toss
Growth	20	Real Value of savings and investments increased
Diversification	15	Money spread across more than one option
Trust Score	10	Loans repaid on time in the Friends' Room

Part	Points	What it checks
Scam safety	10	Scams avoided or spotted

A score of 60 or more unlocks the next chapter. Below that, the student replays the chapter with a tip on their weakest area.

The Life Report

At the end of every chapter:

1. **What you did well:** 2 or 3 specific wins.
2. **What could be better:** 1 or 2 specific misses, explained simply.
3. **What if:** a side-by-side replay showing one changed decision. For example: *"If you had started a ₹500 monthly SIP instead of keeping cash, you would have about ₹X more today, based on the game's assumptions."*
4. **Gyaan Check:** 3 short scenario questions to confirm the lesson.

Rewards

- Badges: Scam Buster, Patient Investor, Trusted Lender, Debt Free, Diversifier
- Class leaderboard ranked by Financial Health Score, never by total wealth

How Toss Life solves the four pillars

Each pillar is taught by doing, not reading, and each is measured inside the game.

Pillar	What students do in Toss Life	How learning is measured
Plan & budget (needs vs. wants, choices, goals)	Split income across tasks with due dates; choose between a friend's gift and a savings goal; set chapter goals	Tasks completed on time; goals reached

Pillar	What students do in Toss Life	How learning is measured
Save & manage (habits, spending, unexpected needs)	Build an emergency fund before the toss; handle bad tosses without borrowing; borrow and repay in the Friends' Room	Safety net score; Trust Score; debt avoided
Understand investing (risk, time, compounding, diversification)	Unlock and compare FD, SIP, gold, debt and index funds; watch Real Value; pool money in bids; see market tosses over years	Growth and diversification scores; "What if" outcomes; Gyaan Check
Stay financially safe (protect information, spot suspicious offers)	Face scam offers, OTP calls and loan apps at tempting moments; spot repeat scams in new disguises	Scams avoided; Scam Buster badge

How it meets the success criteria

Criterion	How Toss Life meets it
Simple	One loop repeated in every chapter; guided first month for beginners
Relatable	Everyday tasks: phone bills, fees, gifts, weddings, first salary
Persona-led	6-question Money Mindset Check places students into 4 personas that adapt the game
Language-ready	All text stored separately; English, Telugu, Hindi and Urdu; voice narration
Game-based	Choices are the assessment; Health Score, Gyaan Check and the before-and-after Mindset Check
Intuitive UI	Wallet, to-do list and calendar that students already understand

Criterion	How Toss Life meets it
NGO-integrated	Class codes, Friends' Rooms per class, facilitator dashboard and discussion prompts

Impact

Toss Life aims to change how students behave with money, not just what they can define, and it measures that change without collecting personal data.

Impact on students

- **Investing confidence:** students understand why time, SIPs and diversification matter, and when an FD is still the right choice.
- **Safer choices:** students learn to recognise scams and high-interest loans before they meet them in real life.
- **Better habits:** budgeting, saving for emergencies and repaying on time become instincts built through play.
- **Responsibility and trust:** lending to and borrowing from friends teaches that money decisions affect relationships.

Impact beyond the classroom

- Students often explain what they learned to their families, carrying ideas like inflation and scam awareness home.
- NGOs get a ready-to-run, engaging session format with built-in progress tracking.
- The game creates a generation of first-time earners who start investing early and wisely.

How impact is measured

Measure	How it is captured
Knowledge gain	Change in Money Mindset Check answers, start vs. end
Investing understanding	Share of students using SIPs, diversification and an emergency fund by Chapter 3

Measure	How it is captured
Scam awareness	Share of repeat scams spotted after the first one
Engagement	Chapters completed and repeat play
Scale	Students reached through each NGO, starting with about 1,500

Delivery, privacy and safeguards

NGO integration

- Facilitators create a class and share a code; each class can run its own Friends' Rooms.
- A facilitator dashboard shows class-level insights, such as which scam fooled most students or which pillar is weakest.
- Each chapter comes with 3 discussion prompts for a guided session after play.
- Chapters, tasks, events and scams are stored as content files, so NGOs can add new scenarios without coding.

Privacy by design

- No names, phone numbers, income, family or location details are collected.
- Persona is built only from 6 indirect, everyday questions.
- Students use an avatar and nickname; progress is linked to a class code and a random device ID.
- Dashboards show class totals, not individual students.

Language and access

- English, Telugu, Hindi and Urdu, switchable anytime, with icons and voice narration for weaker readers.
- A light web app that installs from a link, works on low-cost Android phones, and plays offline once loaded. Multiplayer syncs when online.

Guardrail: educate, do not recommend

- No real fund, bank or company names.

- Every return is labeled as an assumption, not a promise, and market tosses show that losses can happen.
- The Life Report explains trade-offs; it never tells students what to buy.

Prototype scope for the hackathon

The prototype will show one complete chapter end to end, with multiplayer lending and one auction, plus the full Life Ladder as a map.

What we will build

- Money Mindset Check with all 6 questions and persona result
- Chapter 2 (First Job) fully playable: income, task list, spend / save / invest / lend choices
- Investment options: savings, FD, gold and SIP, with assumed returns and Real Value
- Coin toss for life and market events
- Friends' Room for 3 to 4 players: loan requests, loan offers with interest, Trust Score
- One Opportunity Auction with the Pool Bid and the mutual fund reveal
- 2 scams and the Red Flags card
- Life Report with the "What if" replay
- English plus one regional language

Planned after the hackathon

- All five chapters, including both career paths
- Facilitator dashboard and discussion prompts
- Full translation and voice narration
- Pilot with about 1,500 students through partner NGOs, then scale to more schools and cities

Technology

- Web app (PWA) built with React, works offline on low-cost phones
- Real-time multiplayer rooms using Firebase or Supabase
- Game content in JSON files for easy translation and new scenarios
- A simulation engine for tosses, returns, inflation and interest